

ABSTRACT

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Aligarh district in western Uttar Pradesh is one of the most intensively irrigated agricultural zones of the Gangetic Plain, yet no systematic long-term satellite-based record of its vegetation dynamics existed before this study. Vegetation indices derived from satellite data have been used across India to track crop condition and land cover change, but district-level studies that combine multi-decadal trend detection with time series forecasting and probabilistic assessment remain scarce in the published literature, particularly for this part of northern India. This dissertation extracted 24 years of monthly and annual Normalized Difference Vegetation Index (NDVI) data for Aligarh district from the MODIS MOD13Q1 product (250 m, 16-day composite) using Google Earth Engine, producing five structured datasets covering the training period of 2001 to 2024 and an independent validation dataset for 2025. The analysis applied an integrated multi-method framework that included ordinary least squares regression, the Mann-Kendall non-parametric trend test and Sen's Slope estimator, stationarity testing (ADF, PP, and KPSS), ARIMA and Auto SARIMA time series modelling, STL seasonal decomposition, yearly and monthly zonal statistics across 71,346 MODIS pixels, and Bayesian linear regression using Hamiltonian Monte Carlo sampling. Spatial verification was performed in QGIS to confirm the accuracy of the GEE extraction against the GADM Level-2 district boundary. Annual mean NDVI increased from 0.4185 in 2001 to a peak of 0.4923 in 2023 at a rate of 0.0034 NDVI units per year, confirmed independently by OLS regression ($R^2 = 0.868$, $p < 0.001$), Mann-Kendall testing ($Z = 5.28$, $\tau = 0.775$, $p < 0.001$), and Bayesian regression (posterior slope = 0.003360, 95% credible interval [0.00279, 0.00393], $P(\beta > 0) \approx 1.000$). Monthly analysis revealed a strongly bimodal seasonal cycle, with a February Rabi peak of 0.6718 and an April–May inter-seasonal trough of 0.26 to 0.27, characterised by a seasonal strength (F_s) of 0.941. The ARIMA(1,1,0) yearly model and the Auto SARIMA(2,0,0)(0,1,1)[12] monthly model were validated against 2025 observations withheld entirely from model fitting, producing an annual forecast error of 0.19

percent and a monthly Mean Absolute Percentage Error of 9.9 percent. The convergence of classical, non-parametric, and Bayesian slope estimates — all within 0.0001 NDVI units per year of one another — provides unusually strong cross-method evidence for the identified vegetation improvement trend. The most probable driver is the intensification of groundwater-based tube-well irrigation across the district over the study period, consistent with patterns reported for adjacent western Uttar Pradesh districts. The analytical framework developed here, including the reproducible GEE extraction script, the R modelling pipeline, and the out-of-sample validation design, is directly transferable to other Indian agricultural districts and represents a methodological contribution beyond the district-specific findings.

Keywords: NDVI · MODIS MOD13Q1 · Google Earth Engine · Vegetation trend analysis · ARIMA · SARIMA · Bayesian regression · Mann-Kendall test · STL decomposition · Aligarh District · Uttar Pradesh · Gangetic Plain · Time-series forecasting · Remote sensing.

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CHAPTER 1

I NTRODUCTION